Daniel Farnham

Graduate Software Engineer | C++

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Education

BSc (Hons) Games Computing — First-Class Honours, *Northumbria University* May 2026

BTEC Extended Diploma — Creative Digital Media (Games), DDM September 2023
Newcastle College

Work Experience

C# Backend Software Engineer Intern, *Grape Software* Aug 2026 – Present
- Developed a back-end caching service to queue and ingest Food Standards Agency data upon licence creation, leading to an 11% reduction in user request time
- Optimised database access through batched SQL queries and result caching, reducing data-retrieval time by 61%
- Constructed and optimised a weekly Azure background job to refresh imported data from the latest XML feeds, using stored row hashes to skip unchanged records and avoid redundant SQL updates
- Redesigned the SQL Server schema to centralise shared local-authority data, removing 10 unnecessary columns from business records

Freelance Programmer, *INFINITY27/PROTOGÉ* Aug 2025 – Jan 2026
- Extended an existing C++ combat system with a charge-attack component, combining navigation checks, collision sweeps and animation warping for the shipped title Samsara
- Resolved charge-movement errors on slopes and near ledges, correcting target positioning and collision handling through playtesting and working with senior developers.

Projects

Low-Level 3D Renderer — PS4 Dev Kit / C++ 2025
- Implemented a C++ rendering pipeline using the PS4's low-level graphics API, managing GPU buffers, vertex layouts and PSSL shaders for Phong lighting
- Wrote an OBJ parser to convert mesh files into GPU-ready buffers, resolving indices, and combining position, normal and texture-coordinate data.

PitWall — C# / WPF / OpenF1 API 2026
- Built a C#/WPF Formula 1 replay application that combined asynchronous OpenF1 timing, position and telemetry feeds into a time-addressable timeline, allowing for synchronised playback
- Introduced buffered playback to start replays before the full session download completed, reducing initial loading times by 93%

Eye-Tracking and Telemetry Research— C++/ Tobii API/ Unreal Engine 2025
- Wrote a C++ Unreal Engine plugin using the Tobii Game Integration library, recording gaze coordinates, viewed objects and player movement for immersion research
- Analysed eye-tracking, gameplay telemetry and questionnaire responses across three visual-fidelity levels to evaluate changes in player immersion

Skills

Languages: C++, C#, SQL

Technologies & Tools: .NET, ASP.NET, SQL Server, Dapper, Git, Perforce, Visual Studio